

KALIAKPA, Bulgaria

WMO: 155620

Lat: 43.3636N

Lon: 28.4657E

Elev: 64

StdP: 100.56

Time Zone: 2.00 (EUE)

Period: 94-13

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-8.8	-6.2	-11.8	1.4	-8.4	-9.4	1.7	-5.3	24.4	-1.3	23.8	-1.6	14.4	0	0.567

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.1	29.0	23.4	27.8	22.8	26.9	22.3	24.8	27.7	23.9	26.8	23.0	26.1	4.4	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.8	18.8	27.1	22.8	17.7	26.4	21.9	16.7	25.5	75.7	27.8	71.9	26.8	68.6	26.2	28.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
19.6	17.5	12.5	-11.6	30.9	3.5	1.8	-14.1	32.2	-16.1	33.3	-18.1	34.3	-20.7	35.7		
			-11.9	25.9	3.2	1.1	-14.1	26.7	-16.0	27.3	-17.8	27.9	-20.1	28.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	12.9	2.6	3.2	6.0	10.2	16.0	20.8	23.6	23.8	19.5	14.4	9.5	4.5
	DBStd	8.31	4.65	4.29	3.26	2.72	2.73	2.51	2.09	2.30	2.77	3.73	4.03	4.83
	HDD10.0	818	229	191	127	29	1	0	0	0	0	9	57	174
	HDD18.3	2468	487	424	381	244	81	8	0	1	21	128	266	427
	CDD10.0	1876	1	0	4	34	186	324	422	426	285	146	42	6
	CDD18.3	483	0	0	0	0	8	82	164	168	56	6	0	0
	CDH23.3	2244	0	0	0	0	2	222	943	1006	70	2	0	0
	CDH26.7	266	0	0	0	0	0	11	115	140	1	0	0	0
Wind	WSAvg	5.8	7.2	6.7	6.3	5.3	4.4	4.4	4.8	4.9	5.8	6.2	6.4	7.6
Precipitation	PrecAvg	482	36	24	35	34	42	43	40	34	63	49	39	41
	PrecMax	805	105	67	98	117	92	173	147	91	245	185	94	137
	PrecMin	269	5	3	2	3	10	5	2	4	6	1	5	6
	PrecStd	126	26	16	21	23	24	33	32	28	52	37	23	29
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.2	12.0	14.7	18.9	23.2	28.0	30.2	30.7	26.2	23.3	18.2	14.5
		MCWB	9.3	8.6	9.9	13.9	18.4	22.5	23.2	24.3	21.3	19.9	15.5	12.4
	2%	DB	10.4	10.2	12.8	16.5	21.8	26.5	28.8	29.5	24.9	21.5	17.1	13.0
		MCWB	8.5	7.9	9.1	12.9	17.9	21.6	23.0	24.1	20.3	18.7	14.8	11.5
	5%	DB	9.2	9.0	11.3	14.8	20.8	25.6	27.9	28.1	24.1	20.3	15.7	11.6
		MCWB	7.5	7.2	8.5	11.5	17.4	21.1	22.8	23.1	20.0	17.9	13.9	10.1
	10%	DB	8.2	8.1	10.2	13.7	19.8	24.6	27.0	27.1	23.3	19.3	14.6	10.7
		MCWB	6.9	6.4	8.0	11.0	16.7	20.6	22.3	22.5	19.5	16.9	12.9	9.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.1	9.1	10.7	14.8	19.7	23.9	25.5	27.2	22.6	20.7	16.2	12.7
		MCDB	11.6	10.9	13.5	17.8	22.1	26.7	28.2	29.4	25.0	22.4	17.6	14.1
	2%	WB	8.8	8.2	9.6	13.0	18.7	22.7	24.6	25.4	21.7	19.3	15.0	11.6
		MCDB	10.0	9.8	11.9	15.6	21.0	25.5	27.5	28.3	24.0	21.0	16.7	12.9
	5%	WB	7.8	7.4	8.8	12.1	17.9	21.9	23.8	24.2	21.0	18.2	14.1	10.4
		MCDB	8.9	8.8	10.9	14.4	20.2	24.6	26.9	27.1	23.2	20.1	15.7	11.4
	10%	WB	6.9	6.6	8.1	11.3	17.1	21.1	23.0	23.3	20.2	17.0	13.1	9.4
		MCDB	8.1	7.9	10.0	13.3	19.4	24.0	26.3	26.3	22.6	19.0	14.4	10.5
Mean Daily Temperature Range	5% DB	MDBR	3.9	4.0	4.1	3.9	4.3	4.8	5.2	5.1	4.7	4.2	4.0	4.1
		MCDBR	3.7	4.3	5.4	5.3	5.1	5.6	5.7	5.6	5.1	4.4	4.2	4.1
		MCWBR	2.9	3.0	3.4	3.2	3.7	4.3	4.7	4.7	3.8	3.8	3.1	3.5
	5% WB	MCDBR	3.5	4.1	4.9	4.6	4.7	5.0	5.1	5.0	4.6	4.3	4.1	3.9
		MCWBR	3.0	3.1	3.4	3.2	3.7	4.2	4.8	4.8	3.9	4.0	3.4	3.5
Clear-Sky Solar Irradiance	taub	0.356	0.385	0.415	0.457	0.428	0.430	0.438	0.448	0.429	0.398	0.375	0.358	
	taud	2.333	2.252	2.196	2.093	2.264	2.298	2.286	2.251	2.273	2.329	2.329	2.328	
	Ebn at Noon	754	792	817	810	847	845	833	810	794	770	723	710	
	Edh at Noon	84	107	128	153	133	129	129	129	118	98	84	78	
All-Sky Solar Radiation	RadAvg	1.32	2.08	3.38	4.88	6.21	6.80	6.79	6.07	4.32	2.70	1.59	1.19	
	RadStd	0.20	0.25	0.35	0.47	0.34	0.42	0.36	0.28	0.36	0.25	0.21	0.19	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (19 neighbors)	+0.47	N/A	N/A	N/A	N/A	N/A	N/A	-114	+99	+58

Nomenclature: See separate page